

Lab 3: Operating System Installation – Linux OS

Lab Date: June 9th, 2026 **Deadline:** June 16th, 2026 **Instructor:** Shishir Ghimire

Objective

To understand the basics of Operating System installation and basic networking commands by installing VirtualBox and Ubuntu OS using VirtualBox / VMware.

Theory

a. Linux OS

Linux is a free and open-source operating system kernel first developed by Linus Torvalds in 1991. It is based on the Unix operating system and follows the Unix philosophy of modularity and multi-user support. Linux is widely used in servers, desktops, embedded systems, and mobile devices — for example, Android is Linux-based. It uses a monolithic kernel architecture where core OS functions such as memory management, process scheduling, and device drivers run in kernel space.

Key Features of Linux:

- Open-source and freely available under the GNU General Public License (GPL)
- Supports multi-user and multi-tasking environments
- High stability, reliability, and security
- Supports a wide range of hardware architectures
- Large community support and regular updates
- Available in many distributions (distros) such as Ubuntu, Fedora, Debian, Arch Linux, etc.
- Highly customizable — users can modify source code as per requirement

b. Ubuntu OS

Ubuntu is one of the most popular Linux distributions, developed and maintained by Canonical Ltd. It is based on Debian Linux and was first released in October 2004 by Mark Shuttleworth. Ubuntu is designed to be user-friendly, making it an ideal choice for beginners transitioning from Windows or macOS. The name 'Ubuntu' is derived from the Zulu/Xhosa word meaning 'humanity to others'.

Key Features of Ubuntu:

- Based on Debian Linux with APT (Advanced Package Tool) for software management
- Uses GNOME as its default desktop environment
- Released in Long-Term Support (LTS) versions every 2 years, supported for 5 years

- Comes pre-installed with applications like Firefox, LibreOffice, and GIMP
- Free to download, use, and distribute
- Strong community support and extensive official documentation
- Widely used for software development, server deployment, and education

c. VirtualBox / VMware

VirtualBox:

VirtualBox is a free, open-source virtualization software developed by Oracle Corporation. It allows users to run multiple operating systems simultaneously on a single physical machine by creating Virtual Machines (VMs). VirtualBox supports a wide variety of host and guest operating systems, making it suitable for testing, development, and learning.

Key Features of VirtualBox:

- Free and open-source under GNU GPL license
- Cross-platform — runs on Windows, macOS, and Linux hosts
- Supports many guest OSes including Linux, Windows, Solaris, etc.
- Provides snapshot feature to save and restore VM states
- Supports shared folders and clipboard between host and guest
- Virtual networking with NAT, Bridged, and Host-Only adapter modes
- Easy to configure with a graphical user interface

VMware:

VMware is a commercial virtualization platform developed by VMware Inc. (now part of Broadcom). VMware Workstation Pro (for desktop use) and VMware Workstation Player (free version) are commonly used for running virtual machines on personal computers. VMware is known for its enterprise-grade performance and reliability.

Key Features of VMware:

- High performance and stability with advanced hardware support
- Supports Unity Mode to run guest apps on host desktop seamlessly
- Advanced virtual networking options and VLAN support
- VMware Tools for improved integration between host and guest
- Supports 3D graphics acceleration in virtual machines
- Widely used in enterprise and professional environments

Comparison: VirtualBox vs VMware

Feature	VirtualBox	VMware Workstation Player
Cost	Free & Open Source	Free (Player) / Paid (Pro)
Performance	Good	Better
Ease of Use	Beginner Friendly	Slightly Advanced
Snapshot Support	Yes	Yes (Pro only for free)
3D Graphics	Limited	Better Support

Platform	Windows, Mac, Linux	Windows, Linux
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Conclusion

In this lab, we successfully installed Ubuntu OS on VirtualBox as a virtual machine. Through this exercise, we learned the step-by-step process of OS installation in a virtualized environment. VirtualBox provided an ideal sandbox to practice OS installation without affecting the host machine. We also gained an understanding of the Linux/Ubuntu OS structure and its advantages over other operating systems. This lab enhanced our practical knowledge of virtualization technology and Linux-based systems, which are widely used in networking and server environments.

References

1. Course Presentation – <https://docs.google.com/presentation/d/1q1FvwA8yrdd26G5eky-Ilj5avKpkn61E/>
2. VirtualBox Official Downloads – <https://www.virtualbox.org/wiki/Downloads>
3. Ubuntu ISO Drive – <https://drive.google.com/file/d/1RtryMWjRydXCgyePg7rFP657s3fWp7N0/view>